

# Skill Arena - Phase 3 Document 1

## Treasury, Wallet & Ledger Architecture Specification

### Document Purpose

Define the financial backbone of Skill Arena including treasury architecture, wallet systems, token accounting, reserves and ledger design.

### Business Objective

Ensure every token is auditable, every liability is backed and every transaction is traceable.

### Financial Philosophy

Every token must have an owner. Every balance must reconcile. Every liability must be treasury-backed.

### Core Financial Components

Treasury Engine, Wallet Engine, Ledger Engine, Reconciliation Engine, Audit Engine and Compliance Engine.

### Treasury Structure

Player Funds Reserve, Revenue Reserve, Season Reserve, Championship Reserve, Jackpot Reserve and Emergency Reserve.

### Player Funds Reserve

Represents liabilities owed to players. Must remain fully backed and segregated from operating funds.

### Revenue Reserve

Stores platform earnings from fees, premium features and future monetization.

### Season & Championship Reserves

Dedicated funds used for seasonal rewards, tournaments and championship payouts.

### Emergency Reserve

Protected reserve used for operational continuity and risk mitigation.

### Wallet Architecture

Each account contains Live Wallet, Demo Wallet, Pending Balance and Locked Balance components.

### Live Wallet

Real-value token balance available for gameplay and withdrawals.

### Demo Wallet

Practice-only balance with no monetary value.

## **Locked Balance**

Funds temporarily reserved for active matches, tournaments or compliance review.

## **Ledger Architecture**

Immutable double-entry accounting system recording every movement of value.

## **Ledger Entries**

Transaction ID, Timestamp, User ID, Source Account, Destination Account, Amount, Type and Status.

## **Double Entry Rules**

Every debit must have a matching credit entry. No transaction may exist without balancing records.

## **Transaction Categories**

Deposits, Withdrawals, Rewards, Entry Fees, Tournament Fees, Adjustments, Refunds and Administrative Actions.

## **Treasury Reconciliation**

Continuously compares ledger balances against treasury reserves and payment providers.

## **Deposit Processing**

Payment Confirmation → Ledger Entry → Wallet Credit → Treasury Update → Audit Record.

## **Withdrawal Processing**

Request → Validation → Compliance Review → Ledger Reservation → Treasury Validation → Payout.

## **Financial Controls**

Negative balance prevention, reserve validation, liability monitoring and payout limits.

## **Audit Architecture**

Permanent storage of financial events, approvals, adjustments and reconciliations.

## **Compliance Integration**

Supports KYC, AML, transaction monitoring and regulatory reporting requirements.

## **Security Requirements**

All balances calculated server-side. No client-side financial calculations permitted.

## **Performance Requirements**

Financial operations processed reliably with full auditability and reconciliation.

## **Acceptance Criteria**

System can explain every token, every balance, every reserve and every liability at any point in time.